Alessandro Pani Patiño

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Summary

Computer Science student at the University of Costa Rica who builds and ships production software. Replaced a paper-based process at UCR with a web application now used by 50+ staff, and has built and supported websites for 10+ freelance clients since 2022. Available full-time in 2027.

Education

University of Costa Rica | B.S. Computer Science (Software Eng.) *Coursework ends Dec 2027*

Experience

Research & Development Assistant | Ensamblaje Piñero, UCR *Jun 2026 - Present*

- Sole developer on a UCR language and discourse research project, implementing the analysis design defined by a linguistics PhD: built Python pipelines that collected, extracted and segmented 7,063 Legislative Assembly session transcripts (~17 GB) into a traceable corpus.
- Building a benchmark of Spanish part-of-speech taggers, from pretrained pipelines (Stanza, spaCy) to a fine-tuned transformer language model (BETO), with a strict EAGLES evaluator and 67 automated tests.
- Prepared SLURM jobs to fine-tune BETO on NVIDIA A100 GPUs in UCR's high-performance computing cluster, with reproducible environments.
- Building the project's bilingual website (Astro, TypeScript) with CI deployments to Vercel, due Dec 2026.

Full-Stack Developer | SIGECAT, UCR Vice-Rectorate of Administration *Jan 2026 - Jun 2026*

- Replaced paper-based work-schedule and shift declarations with a production web application used by 50+ university staff; built the React + TypeScript frontend, PHP REST API and Oracle data layer.
- Cut the response time of slow Oracle queries from about 4 s to under 1 s, and verified the API with endpoint and concurrency tests.
- Secured the API with token authentication and refresh-token rotation, bcrypt, per-IP rate limiting, account lockout and prepared statements; shipped with Docker and CI (lint, build, PHPStan) on Oracle Cloud.

Freelance Web Developer *2022 - Present*

- Built and maintained websites for 10+ clients in legal services, hospitality and e-commerce, owning requirements, development, deployment and SEO.
- Provide ongoing support: troubleshoot client-reported issues, configure email, DNS, SSL and hosting migrations, and train clients to update their own content.

Software Engineer | deferia.cr, UCR and FAO *Aug 2023 - May 2025*

- Joint UCR project with the UN Food and Agriculture Organization (FAO): turned four years of CNP agricultural price bulletins into a clean dataset of 9,184 records across 56 products (Django, PostgreSQL) and built interactive price-trend charts (React, Plotly).

Project

Museo del Espacio Matemático | Three.js, MediaPipe *Live demo*

- Built 30+ interactive math and physics simulations (hydrogen orbitals, dipole radiation, Fourier epicycles) with WebGL and Web Audio, several controlled by on-device webcam hand tracking.

Skills

Languages: Python, TypeScript/JavaScript, PHP, SQL (Oracle, PostgreSQL, SQLite)
Web: React, Astro, Django, REST APIs
Data and ML: pandas, scikit-learn, Power BI, PyTorch, Hugging Face Transformers, spaCy, Stanza
Tools: Git, Docker, Linux/Bash, SLURM (HPC), pytest, CI pipelines, Jira, Oracle Cloud Infrastructure
Spoken: Spanish and Italian (native), English (C1, professional)